

**RAMCO INSTITUTE OF
TECHNOLOGY**

STRUCTA
ANNUAL
TECHNICAL
MAGAZINE
2022-2023

VOL -3

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MAGAZINE

2022-2023

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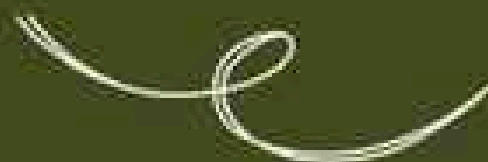
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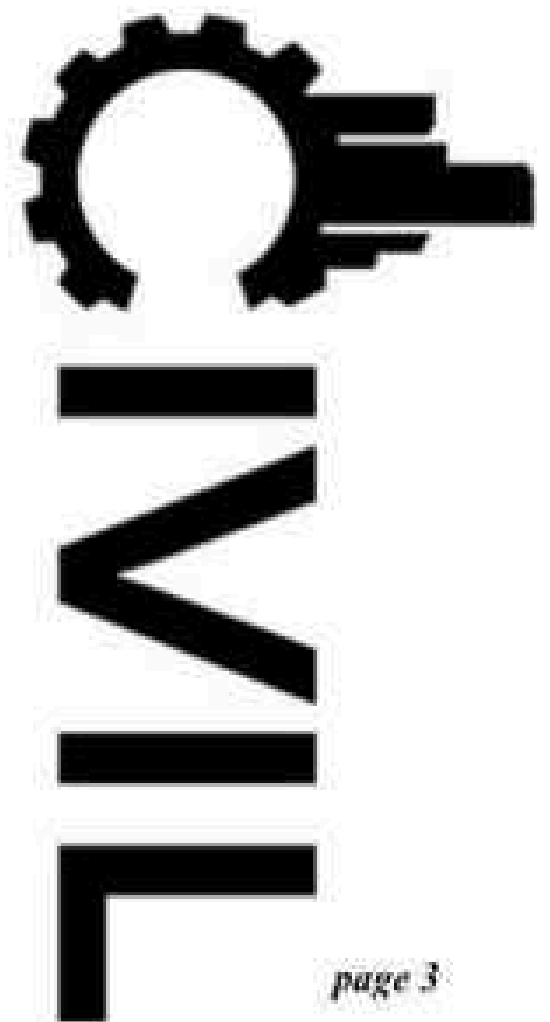
CIVIL ENGINEERING



DEPARTMENT OF CIVIL ENGINEERING

About the Department

The Department of Civil Engineering was established in the year 2013. At present, it offers undergraduate programme, B.E. in Civil Engineering with a sanctioned intake of 60 students. The Department broadly covers important fields such as Surveying, Geotechnical Engineering, Structural Engineering, Hydraulics and Water Resources Engineering, Construction Materials and Practices, Environmental Engineering and Transportation Engineering.



The Department has well established modern laboratories to cater to the needs of academic and research activities. The laboratories of the department include Concrete and Highway Engineering Laboratory, Surveying Laboratory, Environmental Engineering Laboratory, Soil Mechanics Laboratory, Hydraulic Engineering Laboratory, Strength of Materials Laboratory, CAD Digitalized Construction Laboratory, RIT - Centre of Excellence for Building Information Modelling (BIM), RIT ICT Academy Centre of Excellence for Design Powered by Autodesk, Construction Practices Laboratory, Centre for Geospatial Technology, Project & Consultancy Laboratory.

Our Department has Anna University recognized R&D Centre from the month of January 2024 which enables us to recruit and guide PhD scholars.

The Department is very active in Industrial collaboration by signing MoU with prominent organisations like Land Coordinates Technology, L&T EduTech, TechApps Consulting, Pyramid IAS Academy, Builders Association of India (BAI), V-Invent Chemilab Private Limited, Ramco Cements Ltd., etc., to bridge the gap between academia and industry. Through initiatives like guest lectures, conferences, seminars, webinars, industrial and site visits, as well as regular value-added courses, the Department ensures students remain updated on recent advancements in Civil Engineering. Additionally, a strong industry-institute interaction is maintained by providing consultancy and testing services in various domains.

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The Department encompasses professional societies such as Indian Concrete Institute (ICI), Institution of Engineers IE(I) India, Indian Chapter of American Concrete Institute (ACI), Indian Green Building Council (IGBC) which provides students a platform to acquire knowledge in various technical activities. The Civil Engineering Association RIT STRUCTA serves as a platform for students to make technical presentations, gain knowledge about recent advancements and bring together novel ideas.

Vision

◆
Create knowledgeable and professionally competent Civil Engineers who are capable of providing global solutions to the problems related to Civil Engineering.

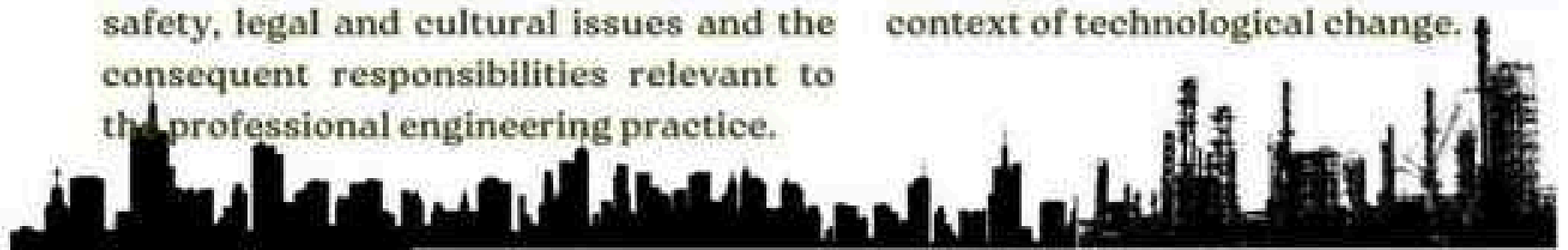


Mission

- Impart high quality education and technical skills to transform students into Professional Civil Engineers.
- Establish state-of-the-art infrastructure for updating knowledge in recent developments and cutting edge technology.
- Bridge the gap between academics and industry by practical training and providing solutions for design projects related to Civil Engineering.
- Motivate the faculty members and students in the teaching learning process for achieving excellence to meet the global standards.
- Produce graduates with high professional and ethical standards for fulfilling the needs of the industry and society.

Programme Outcomes(POs)

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

- Graduates of the programme will be professionally competent and apply the fundamental concepts of mathematics, science and Civil Engineering to solve problems in Civil Engineering.
- Graduates of the programme will perform their responsibilities as professional Civil Engineers with quality and ethics.
- Graduates of the programme, as part of an organization or as Entrepreneurs, will continue to learn with evolving technologies.
- Graduates of the programme will innovate and provide global solutions for a sustainable built environment.
- Graduates of the programme will contribute to the development of sustainable Infrastructure for the betterment of society.

Program Specific Outcomes (PSOs)

On completion of the degree, the students will be able to:

- Communicate and present civil engineering projects effectively.
- Use the techniques, skills, and modern engineering tools necessary for civil engineering practice and project management.
- Provide sustainable solutions to civil engineering problems.
- Perform as design consultants in construction industry for the design of civil engineering structures.



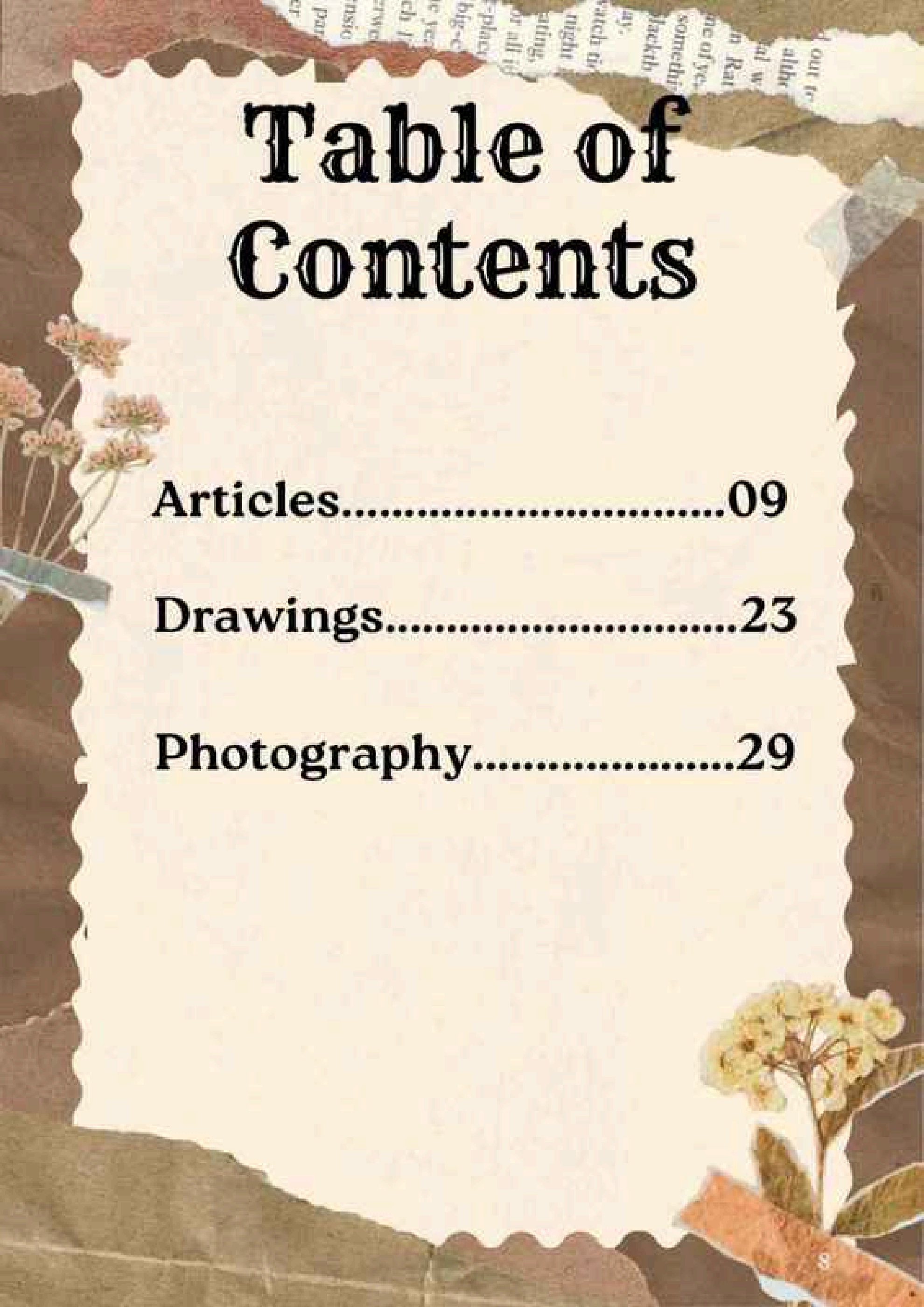


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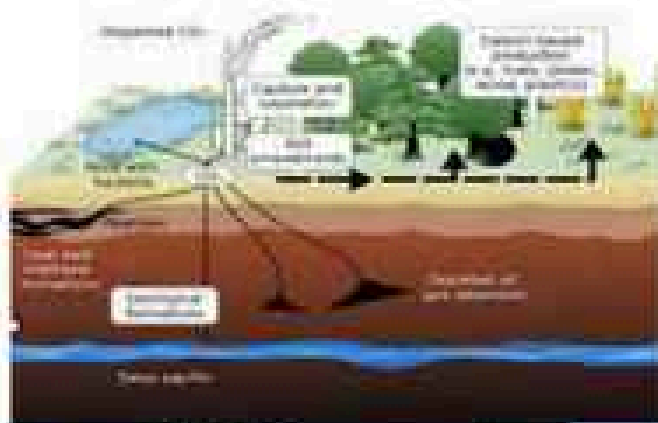
Articles

CARBON SEQUESTRATION



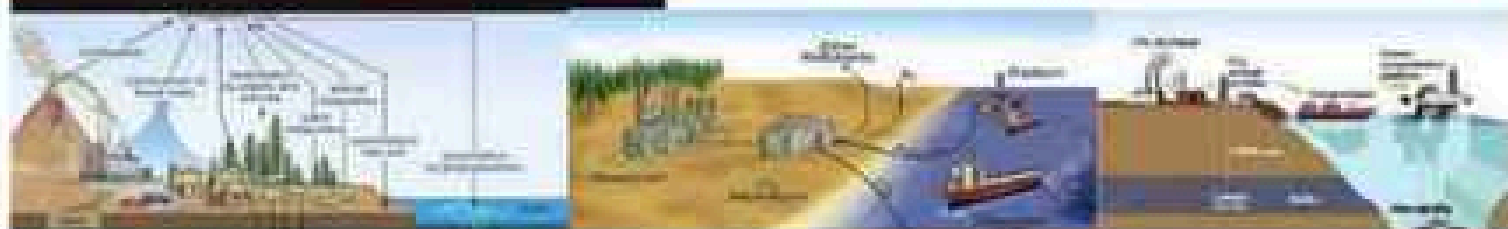
The method of absorbing and storing atmospheric carbon dioxide is known as carbon sequestration. It is one method of decreasing carbon dioxide levels in the atmosphere with the purpose of slowing global climate change.

Carbon sequestration occurs naturally, but it can also be increased or done by technology, such as in carbon capture and storage projects. Carbon sequestration can be classified into two types: geologic and biotic (also known as bio sequestration).



What is Carbon Sequestration ?

Tackling Global Warming



CO₂ is spontaneously taken from the atmosphere via biological, chemical, and physical processes. Changes in land usage and agricultural practices, such as switching crop land to land for non-crop fast growing plants, might hasten these changes. Similar effects have been achieved by artificial processes, such as large-scale, artificial capture and sequestration of industrially produced CO₂ using deep saline aquifers or ageing oil fields. Other carbon sequestration solutions include bio-energy with carbon capture and storage, biochar, improved weathering, and direct air injection.

Other carbon sequestration solutions include bio-energy with carbon capture and storage, biochar, improved weathering, and direct air carbon capture and sequestration. Forests, kelp beds, and other plant life absorb and bind carbon dioxide from the atmosphere as they grow. However, because long-term sequestration cannot be guaranteed, these biological reservoirs are regarded as volatile carbon sinks. Natural disasters, like as wildfires or disease, economic constraints, and shifting governmental agendas, for example, can cause sequestered carbon to be released back into the atmosphere. Carbon dioxide taken from the atmosphere can also be stored in the Earth's crust as insoluble carbonate salts (mineral sequestration) or by injecting it into the subsurface. These technologies are called non-volatile because they extract carbon from the atmosphere and sequester it indefinitely and for an extended period of time (thousands to millions of years).

The following technologies have been proposed to improve carbon sequestration processes in seas, although none have seen widespread application: Seaweed farming, ocean fertilisation, artificial upwelling, basalt storage, mineralization, and deep sea sediments, as well as the addition of bases to neutralise acids, are all examples. The direct deep-sea carbon dioxide injection concept has been abandoned.

Dr.M.Indhumathi
AP/civil



Climate Change Impact in Future India: Challenges and Opportunities



Introduction:

India, the world's second-most populous country, faces significant challenges in the wake of climate change. As global temperatures continue to rise and extreme weather events become more frequent, India is particularly vulnerable to the adverse effects of climate change. This article explores the potential impacts of climate change on future India, highlighting the challenges it poses and the opportunities for adaptation and mitigation.



1. Rising temperatures and heatwaves:

India is projected to experience an increase in average temperatures, leading to more frequent and intense heatwaves. High temperatures can have severe implications for public health, agriculture, and water resources. Heat-related illnesses, reduced crop yields, and increased water stress are major concerns. However, innovative strategies such as heat-resilient infrastructure, urban greening, and early warning systems can help mitigate the impacts of rising temperatures.

2. Changing rainfall patterns and water scarcity;

Climate change is expected to disrupt India's monsoon system, resulting in changing rainfall patterns. Some regions may experience more frequent and intense rainfall, leading to floods and landslides, while others may face prolonged droughts and water scarcity. Efficient water management practices, including rainwater harvesting, watershed management, and water-efficient irrigation techniques, are crucial for adapting to changing rainfall patterns and ensuring water security.

3. Sea-level rise and coastal vulnerability;

India's extensive coastline is susceptible to the impacts of sea-level rise. Coastal erosion, saline intrusion into groundwater, and increased vulnerability to storm surges pose significant challenges. Protecting coastal ecosystems, implementing coastal defense measures, and adopting resilient urban planning strategies can help mitigate the adverse effects of sea-level rise and protect vulnerable communities.

4. Agricultural impacts and food security;

Agriculture, a vital sector in India, is highly sensitive to climate change. Changing rainfall patterns, increased frequency of extreme weather events, and pest and disease outbreaks pose significant threats to crop production. Diversification of crops, adoption of climate-smart agricultural practices, investment in agricultural research and development, and improved access to information and technology can enhance resilience and ensure food security in the face of climate change.

5. Health risks and disease outbreaks;

Climate change can exacerbate health risks in India, including the spread of vector-borne diseases, heat-related illnesses, and air pollution. Increased temperatures and changing precipitation patterns can create favorable conditions for diseases like dengue, malaria, and cholera. Strengthening healthcare systems, improving early warning systems, and promoting climate-resilient infrastructure and urban planning are essential for safeguarding public health.

Biodiversity loss and ecosystem degradation:

India is known for its rich biodiversity and diverse ecosystems, but climate change poses a significant threat to these invaluable resources. Habitat loss, species extinction, and ecosystem degradation are potential outcomes of climate change. Conservation efforts, sustainable land use practices, and protected area management can help mitigate biodiversity loss and ensure the preservation of critical ecosystems.

Conclusion:

India faces numerous challenges in the face of climate change, with potential impacts on public health, agriculture, water resources, coastal areas, and biodiversity. However, amidst these challenges lie opportunities for adaptation and mitigation. By investing in sustainable infrastructure, implementing climate-smart agricultural practices, promoting renewable energy sources, strengthening institutional capacity, and raising awareness, India can build a more resilient and sustainable future. Collaboration between government, private sector, civil society, and international partners is crucial to effectively address the impacts of climate change and secure a prosperous future for India and its citizens.

Mrs. A. Leema Margret

HOW THE TRAINS ARE NUMBERED?



Indian Railways switched to a system of 5-digit numbers that are supposed to be used for all passenger trains across its system during the year 2010.

The first digit

In the 5-digit train numbering scheme, the first digit indicates the type of the passenger train, as follows:

0 is for special trains (e.g., summer specials, holiday specials, etc.)

1 is for all long-distance trains, including the Rajdhani, Shatabdi, Jan Sadharan, Sampark Kranti, Garib Rath, Duronto, and other classes.

2 is also for long-distance trains; it is to be used when train numbers starting with 1 are exhausted in any series.

3 is for Kolkata suburban trains.

4 is for suburban trains in Chennai, New Delhi, Secunderabad, and other metropolitan areas.

5 is for passenger trains with conventional coaches

6 is for MEMU trains

7 is for DMU (DEMU) and railcar services.

8 is currently reserved

9 is for Mumbai area suburban trains

The Second digit indicates the railway zone



Third digit identifies the railway zone where the train is maintained, except for 0

The fourth and fifth digits are random numbers, making each five-digit train number unique

Reference:

1. https://en.wikipedia.org/wiki/Train_numbering_in_India



Mr.T.Chockalingam.

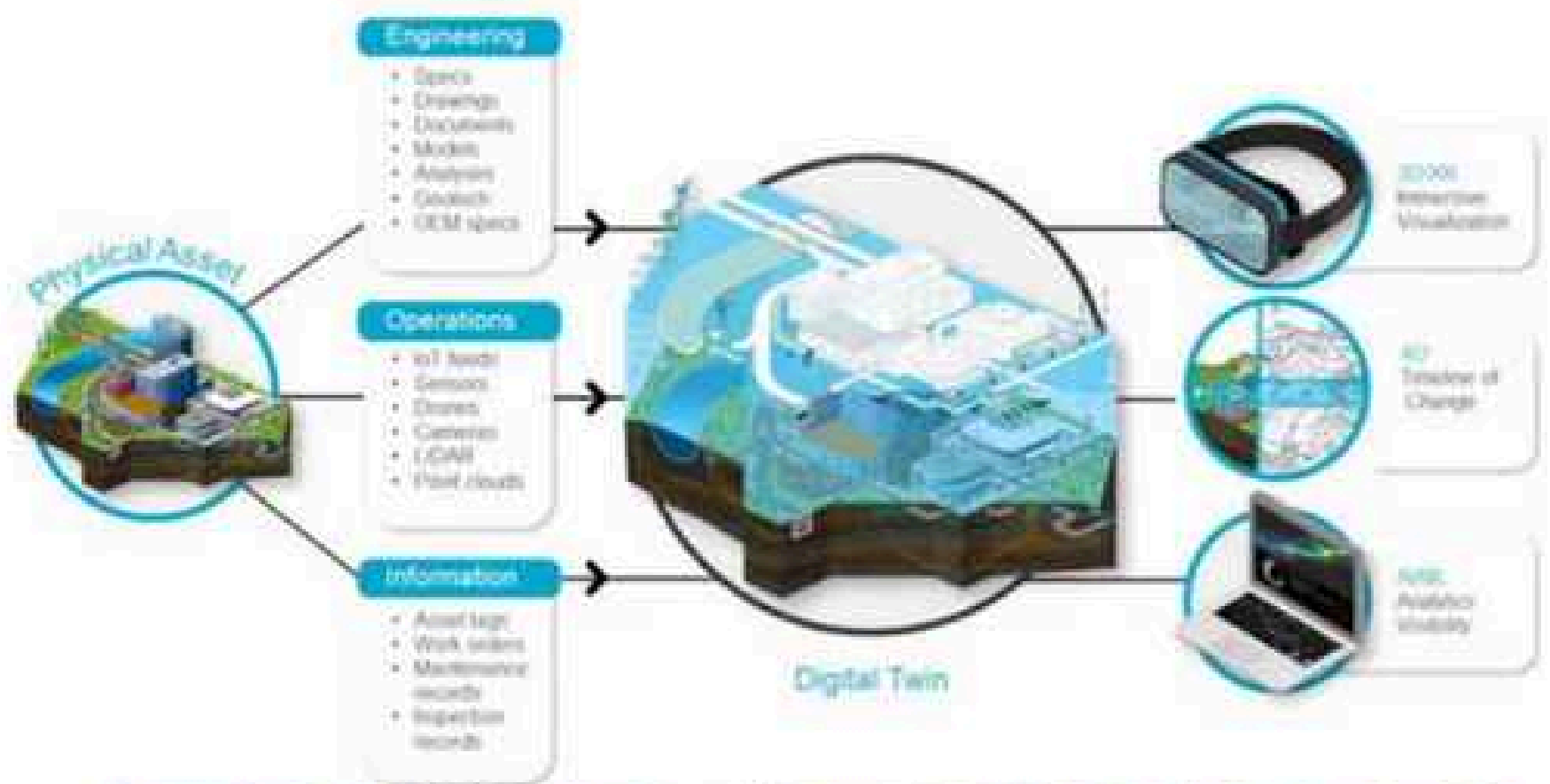
AP(SG)/Civil

Digital Twins



Dr. G. Karthikeyan
AP(SG)/Civil

According to the Digital Twin Consortium, a digital twin is “a virtual representation of real-world entities and processes,” where the representation is kept in-sync with the real world. Those entities and processes may form a real-world system or network. The virtual representation is a complex model consisting of multiple simpler models and simulations woven together with a digital thread. Digital twins can deliver critical insights into the outcomes that may result from various real-life decisions before they are taken. This provides the ability to improve decision-making by identifying the most beneficial action. The fidelity of a digital twin is tied to the accuracy and timeliness of observations of connected assets from the real-life system. Interventional mechanisms can react to changes in the virtual representation and make corresponding changes in the real world. These interventions can involve a human-in-the-loop by sending an alert to a worker, or they can be automated via IoT or traditional control systems. While this level of maturity is still the minority in digital twin use, it is anticipated to grow in the coming years. The Digital Twin Consortium also emphasizes that digital twins are implemented in digital twin systems. The virtual representation is only one part of those systems. A practical digital twin system also requires visualization, analytics, security, data integration, automation, and observational and interventional mechanisms that keep the virtual and real worlds in sync. Intelligent transportation and the digital model are the foundation of a digital city. Digital twins provide a lightweight model, which expands the use of BIM applications beyond the project to the delivery of digital assets.



For the infrastructure industry, digital twins can be used throughout an asset's lifecycle:

- Planning: site surveys and reality capture or 3D mesh;
- Design: stakeholder engagement, planning and visualization, quality review, and change tracking;
- Analysis: pedestrian, flood, and other simulations;
- Construction: logistics simulation, project progress, status review, and 4D modelling;
- Operations and Maintenance: health, safety, social, and environmental (HSSE) training, operators training, maintenance planning, shutdowns, action verification, remote inspection, leak detection, and corrosion detection.



SOIL CEMENT

Soil-cement (SC) is an engineered, densely compacted mixture of soil/aggregate, portland or blended cement, other cementitious materials (possibly), and water. SC is known by a variety of names including cement-stabilized base, cement-treated aggregate base, cement-treated soil, and even dirtcrete. Regardless of what it is called, the principles governing its composition and construction are the same. There are a wide variety of applications of SC, and all have the common advantages of using SC. On-site soils are amended to achieve the desired physical properties to build long-lasting structures that are economical and have a low environmental impact over their life span.

Applications

SC is used in numerous construction applications for pavements, water resources, and geotechnical projects.



**"IF YOU WANT TO
HAVE FOOD, CONSERVE
SOIL DUDE"**



pavements applications include:

- Base courses for streets, roads, highways, shoulders, airports, and parking and storage areas.
- Subbase courses for rigid pavements.
- Modification or stabilization of subgrade soils.

Water resources applications include:

- □ Slope protection for embankment dams, levees, and streambanks.
- □ Liners for impoundments and channels.
- □ In-stream grade control structures.

Geotechnical applications include:

- □ Modification or stabilization of foundation soils
- □ Slurry cutoff walls
- □ Deep mixing methods

Engineering Properties:

There are four major variables that control the engineering properties of SC materials:

1. the characteristics of the soil
2. the proportion of cement in the mix
3. moisture conditions (water content)
4. the degree of compaction

The "soil" material in SC can be almost any combination of sand, silt, clay, gravel, or crushed stone. Local granular materials, such as slag, caliche, lime rock, and scoria, plus a wide variety of waste materials including cinders, fly ash, foundry sands, and screenings from quarries and gravel pits, can all be utilized as soil material. The PCA Soil Primer has basic information on soils regarding their influence on the design, construction, and performance of SC as well as considerable information on soil survey and sampling methods.

Since SC is a structural material, it possesses engineering properties of a magnitude dependent primarily on type of soil, cement content, degree of compaction/consolidation, curing conditions, and age.



By

Mrs. D. Darling Helen Lydia

AP/Civil

TECHNICAL FEATURES OF THE CONSTRUCTION OF HIGH-RISE BUILDINGS

Abstract. High-rise buildings are unique construction objects, in the design and construction of which special rules and regulations apply. The purpose of the work is, through an analysis of the literature and technical documentation, to present the main features of the construction of high-rise buildings. Based on the results of the study, differences in the concept of "high-rise building" in different countries were identified, specific problems encountered in high-rise construction were described, and design features of high-rise buildings were described. Special attention is paid to additional difficulties arising in the construction of high-rise buildings; due to the specificity of loads, construction technology, operation of high-rise buildings, and other reasons.

Introduction

construction. Tall buildings have features that significantly distinguish them from High-rise buildings are being built in connection with the growing population of cities, lack of land, as a rule, these are prestigious projects of special architectural significance, reflecting the state of the scientific and technical process, innovative technologies in ordinary buildings. These are prestigious projects of special architectural significance, reflecting the state of scientific and technological progress, innovative technologies

The first high-rise buildings were built in the USA (Chicago). The increase in land prices, technical progress in building and construction and the reliable construction of the elevator by the American engineer E. Otis led to the appearance of the first many-story houses, which were called "skyscraper".



At the turn of the 19th and 20th centuries, large volumes of office construction, as well as requirements for high concentration and building density, led to an increase in the number of storeys in buildings. At the same time, in addition to the possibility of erecting high-rise buildings on small land plots, the prestige and advertising of firms located on them played a significant role. That is why high-rise buildings are sometimes called "prestige buildings", and the architecture of these buildings is called "the big business style"

In Europe, high-rise buildings appeared only in the middle of the 20th century. At first, these were detached buildings in a historic city area: Pirelli in Milan, Montparnasse Tower in Paris, etc. Since the 1960s high-rise buildings are located mainly outside urban historical zones.

In the 1990s the most dynamically high-rise construction is developing in the countries of Southeast Asia. In 1990-2005, 445-450 m high structures were erected in Kuala Lumpur, Shanghai, and Taipei, for example. Petronas skyscraper (Kuala Lumpur, 1995), consisting of two towers (38 floors each) and recognized in 1996 as the tallest building in the world.

The first high-rise buildings in Russia were erected in the late 1940s - early 1950s in Moscow. Seven high-rise buildings were built, the most significant of which is the main building of Moscow State University on the Sparrow Hills (Figure 2). By the beginning of the 21st century, a number of high-rise buildings were erected in Moscow, including administrative buildings (Gazprom, the central office of Sberbank) and residential complexes (Scarlet Sails, Triumph Palace, etc.)

What is now considered a high-rise building? A high-rise building is any building with a height of more than 75 meters, that is, more than 25 floors. But such a standard is used only in Russia. In other countries, this term usually refers to buildings with a height of 35-100 m, and buildings above 100 m are considered skyscrapers. Skyscrapers above 300 meters are called superhigh.

The minimum height of the skyscraper building remains controversial. In the United States and Europe, skyscrapers are considered buildings at least 150 meters high. There is some uncertainty in the classification and ranking of high-rise buildings due to different measurement methods. The generally accepted standards are those developed by The Council on Tall buildings and Urban Habitat (CTBUH), an international organization dedicated to high-rise construction.

There are three criteria for measuring height:

1. height from the sidewalk level to the highest point of the structural elements of the building, not including television radio antennas and flagpoles;
2. to the floor level of the highest available floor of the building;
3. to the highest point (the height of the building to the tip of the antenna / spire).

At the moment, the first criterion is the main one, it is he who is used in compiling the rating of the highest buildings.

Features of the construction of high-rise buildings:

High-rise buildings are construction object of high risk and engineering complexity, which indicates the need for a thorough analysis of decisions made and taking into account the results of the extensive international experience in high-rise construction.

Despite the accumulated world experience in the construction of high-rise building, there are currently no general regulated rules for the selection of structural solutions for load-bearing Systems, enclosing structures and materials for their implementation. In each case, the design engineer makes a technical decision in accordance with the requirements established by international or national standards, design standards or other documents.



Due to the high height, the layout, design and technical equipment of high-rise buildings are qualitatively different from buildings with lower floors. The design of each high-rise building is unique, because it solves the specific problems that arise during high-rise construction:

- Strong influence of environmental factors (wind, noise, air temperature drops);
- Providing fire protection and safe evacuation of people in an emergency;
- Difficulties in the design of ventilation systems, heat supply and heating;
- A large concentration of cable equipment and pipelines stretched in mines to a huge height;
- The need for continuous monitoring of the main load-bearing structures of the building and base soils, etc.

Materials:

Currently, tall buildings are increasingly preferred to be made of reinforced concrete, since this material has the highest fire resistance, and its strength characteristics are close to the strength of steel. The requirements for concrete as a structural building material for high-rise buildings have become especially stringent. Therefore, without modern technologies for the modification of monolithic concrete, which provide the necessary frost resistance, fire resistance, impact resistance and durability under aggressive influence, high-rise buildings cannot do. Concrete grades B80 and B100 were created and used, while the lower grades B60 and B70, high-strength concrete, are used in a wide range of construction practices.

Building foundation

A huge load is placed on the foundation of the building: because the specific pressure reaches 500-800 kPa or more.

The depth of foundation can be 15-25 m.

Nowadays, three types of foundations have been widely used in the design and construction of high-rise buildings: pile, slab and pile-slab. Massive slab foundations are the simplest and most economical design solution.

Structural schemes of high-rise buildings

With the development of high-rise buildings, several design solutions for these buildings have been developed:

1. frameless-structure with parallel bearing walls;
2. core structure with bearing walls;
3. box-shaped structure;
4. frame structure with cantilever slabs in the level of each floor;
5. frame structure with girderless floor slab panels;
6. structure with storey high cantilevers in the level of each second floor;
7. structure with suspended storeys;
8. structure with storey high staggered trusses;
9. framed and-ledged structure;
10. frame-core structure.

The choice of a specific design decision depends on many factors: the height of the building, construction conditions, architectural and planning requirements. Constructive versions of the load-bearing enclosures of multi-storey buildings (box systems). In boxed systems, the outer bearing shells can be made in the form of beveled gratings made of steel or reinforced concrete. The drap-free grille does not create difficulties when placing a translucent fence on the facade of a high-rise building, but is inferior to the diagonal in terms of ensuring structural rigidity.

Bearing elements of structural systems of high-rise buildings

Shelf frame systems-columns, pylons and other similar elements are constructed using the so-called high-strength (HSC) and high-quality (HQC) concrete, the compressive strength of which reaches 100 MPa and higher. The overall dimensions of the columns and the number of service valves are determined by many factors. In case of insufficient bearing capacity, rigidity or longitudinal stability of the frame, reinforced concrete columns with an external steel cover or internal rigid reinforcement are used. This solution allows to increase the fire resistance of the structure.

wall:

The walls of high-rise buildings are made of less durable concrete compared to the concrete used for the construction of columns. In high-rise buildings, the system of load-bearing walls has a monolithic concrete arrangement. This is due to the need to give the frame the maximum possible rigidity, which is technically difficult to provide with prefabricated versions.

Staircase elevator structures

Staircase elevator structures of buildings play a special role in ensuring communication between floors and evacuation of people in case of emergency. Staircase elevator structures can combine the functions of communication lines and evacuation or can be performed separately in both cases, their technical equipment is subject to certain requirements related) to ensuring safety parameters.

Additional difficulties

The specificity of the loads, the technology of construction, operation of high-rise buildings causes a number of additional difficulties

Fire safety, evacuation

Tall buildings, unlike buildings of normal number of storeys, have a high fire hazard due to their specificity.

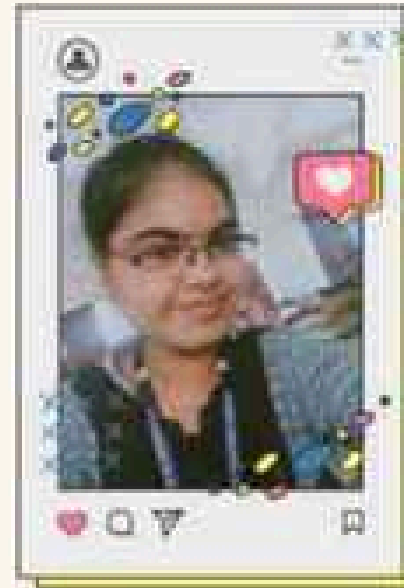
For those who are in skyscrapers, the fire hazard increases due to the fact that there are difficulties in evacuation and rescue. Overlapping evacuation routes with products of combustion and fire is the main cause of the tragic consequences. Combustion products block emergency exits, elevator shafts and stairs. The spread of smoke and toxic gases can reach tens of meters per minute. After a few minutes, the building will be filled with smoke, and it will be impossible to stay indoors without respiratory protection. The smoke of the upper floors, where reconnaissance, rescue and supply of fire extinguishing agents are most difficult, is most intense.

In addition, in the event of a fire elevator equipment and fire protection

Systems often fail

consequences:

- Low fire resistance of building structures and engineering equipment, especially metal beams and trusses;
- Large internal volume without fire barriers
- Lack of a sufficient number of stairs and their necessary width for evacuation;
- The presence of numerous holes in the walls and ceiling for the use of air conditioners, electrical equipment and other technological needs;
- Lack of evacuation plans in the event of an accident or fire;
- The presence of false ceilings;
- A large number of combustible equipment, furniture and cladding.



By

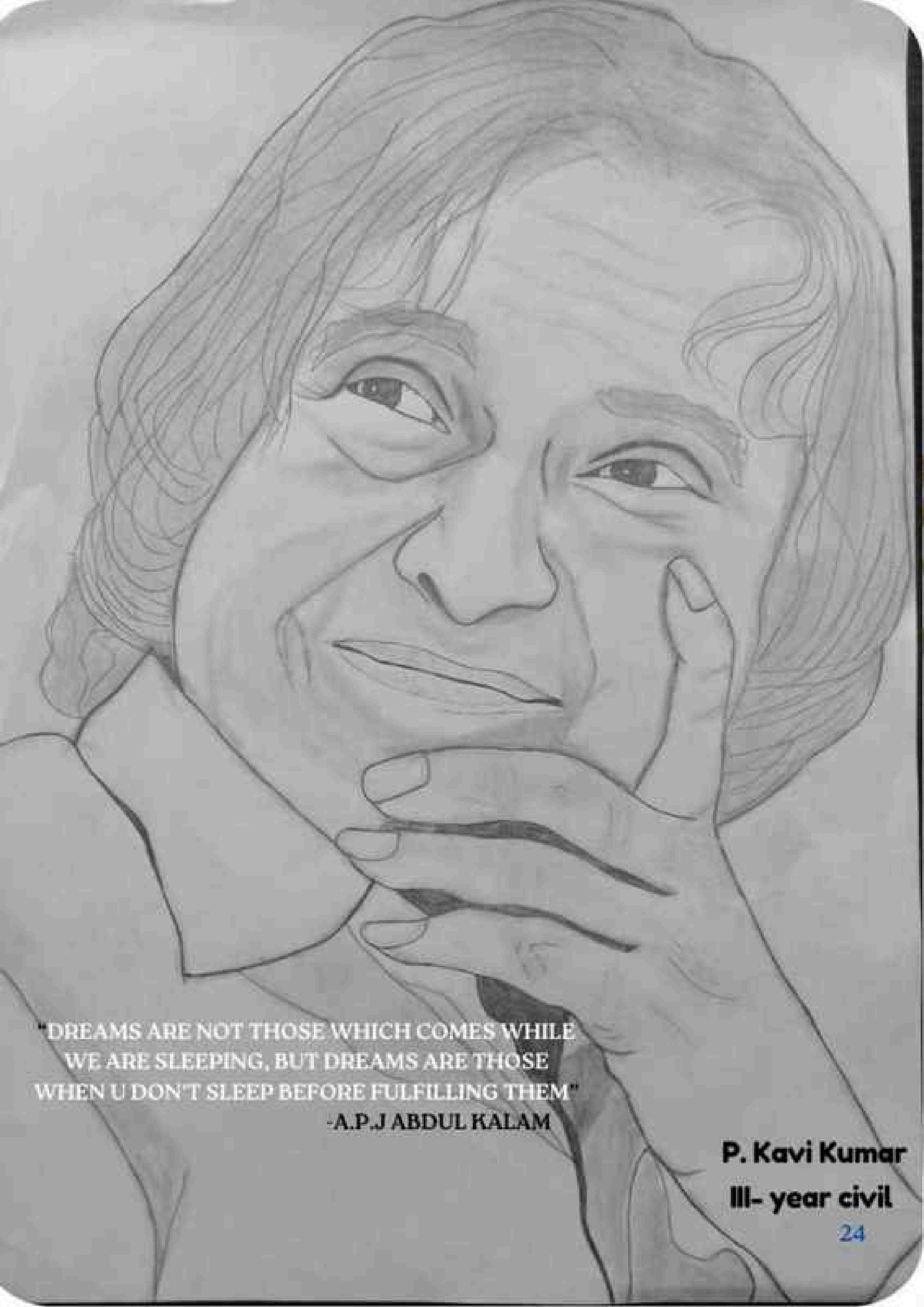
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I-year civil



Art Gallery



Creative Stationery
Lorem ipsum



**"DREAMS ARE NOT THOSE WHICH COMES WHILE
WE ARE SLEEPING, BUT DREAMS ARE THOSE
WHEN U DON'T SLEEP BEFORE FULFILLING THEM"**

-A.P.J ABDUL KALAM

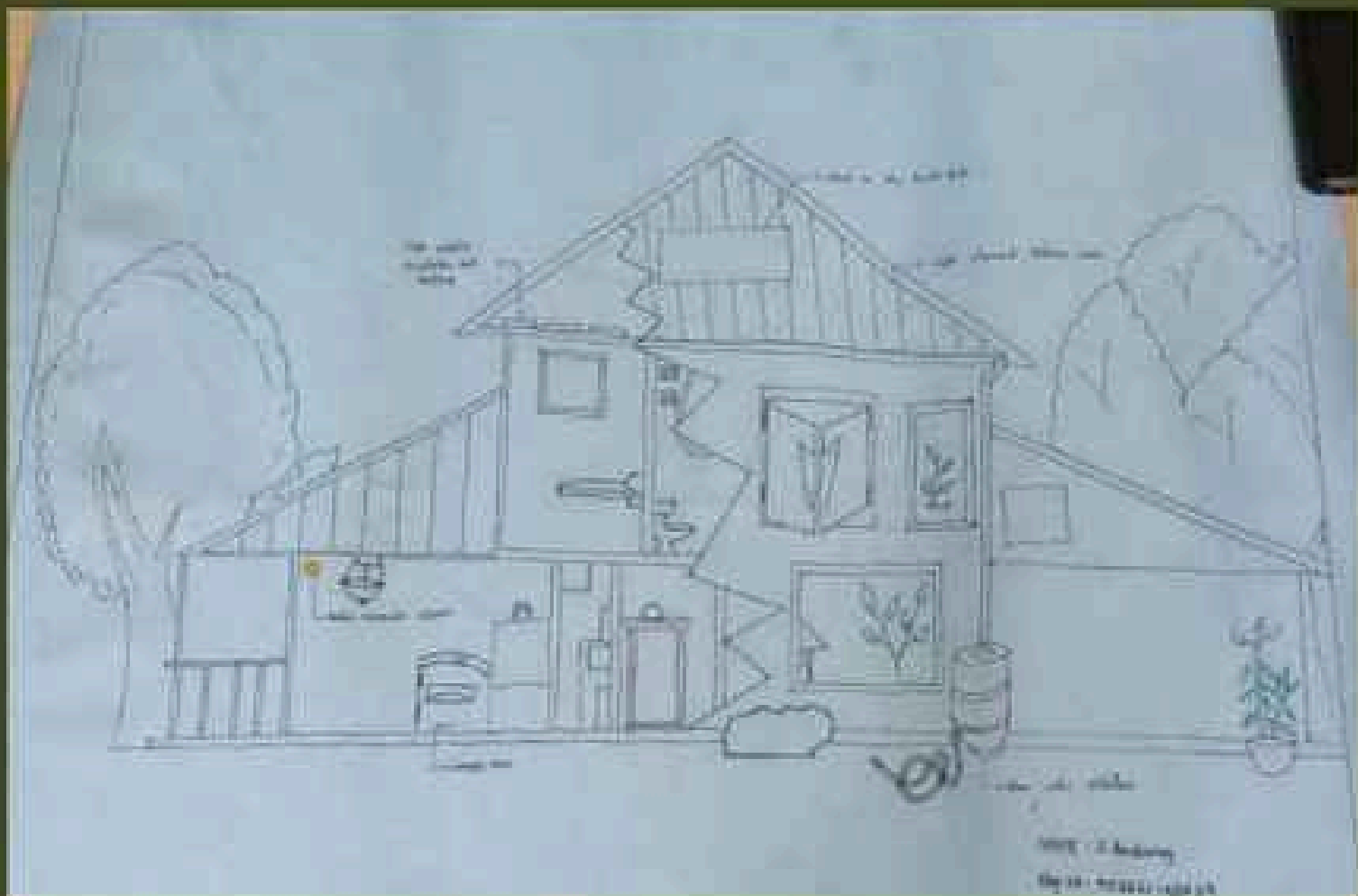
**P. Kavi Kumar
III- year civil**



I don't want to protect
environment. I want to
be a world where the
environment doesn't need
protecting."

P. Sibi Bala
II year civil

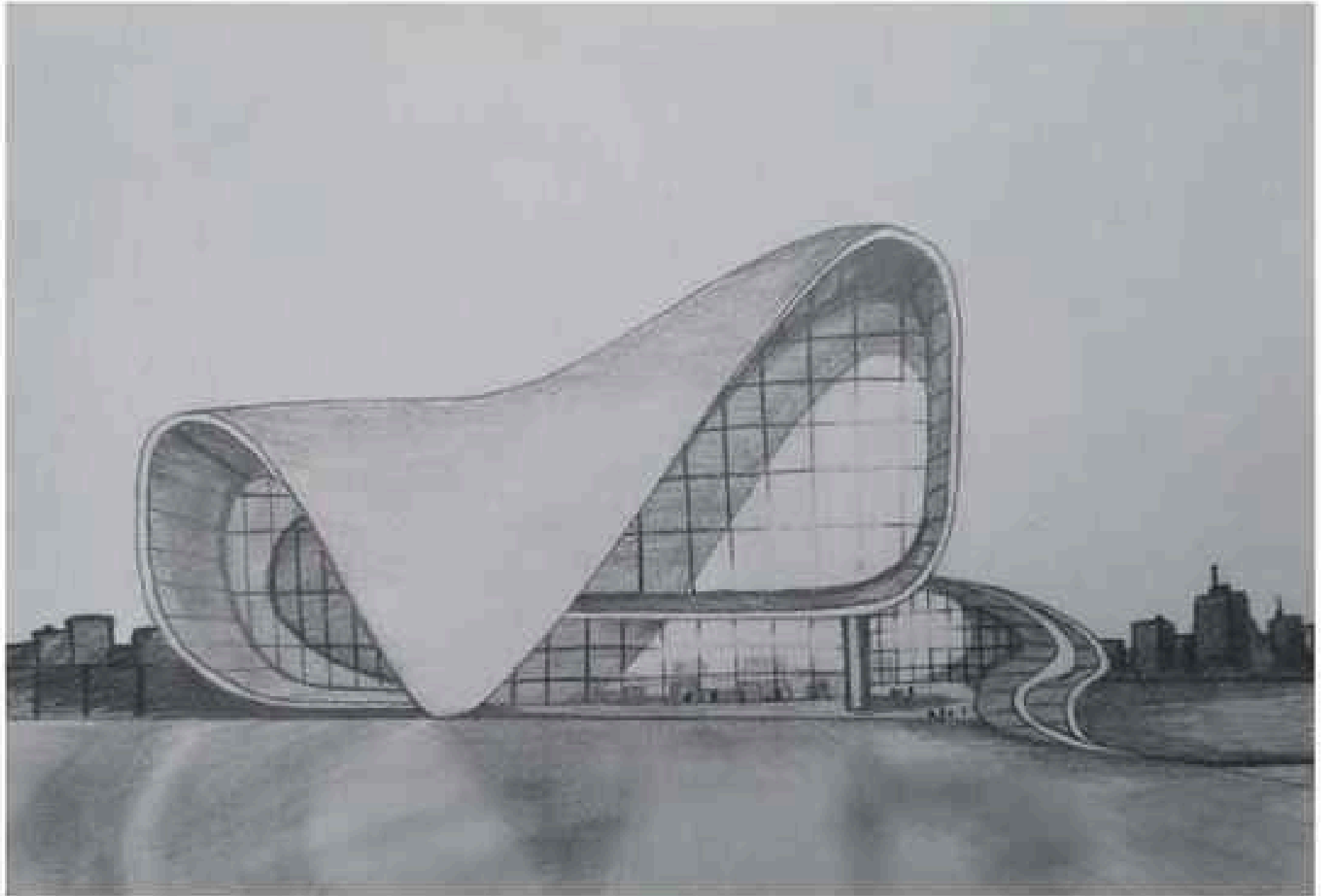
NAME : P. SIBI BALA
REGISTER NO : 153A21103634
Dip / Year : IInd Year / Civil



Mr. Pandiraj
II year civil

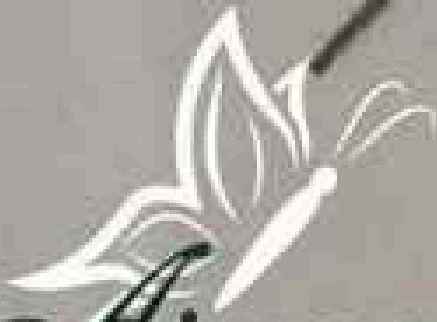
"Green is the fresh emblem of well-founded hopes. In blue the spirit can wander, but in green, it can rest."

Drawings



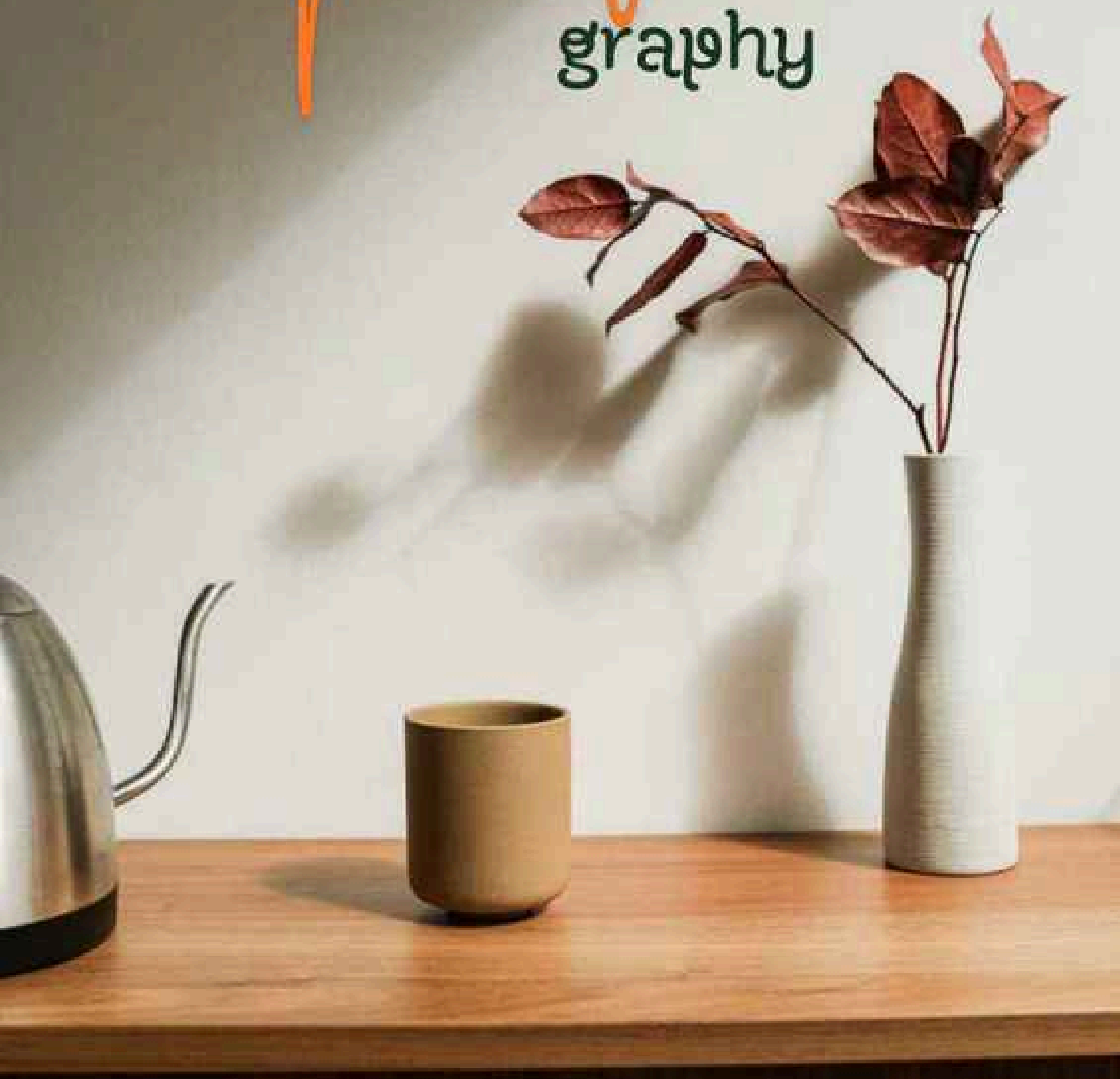
Mr. Vijay Mallaiya
1- year civil

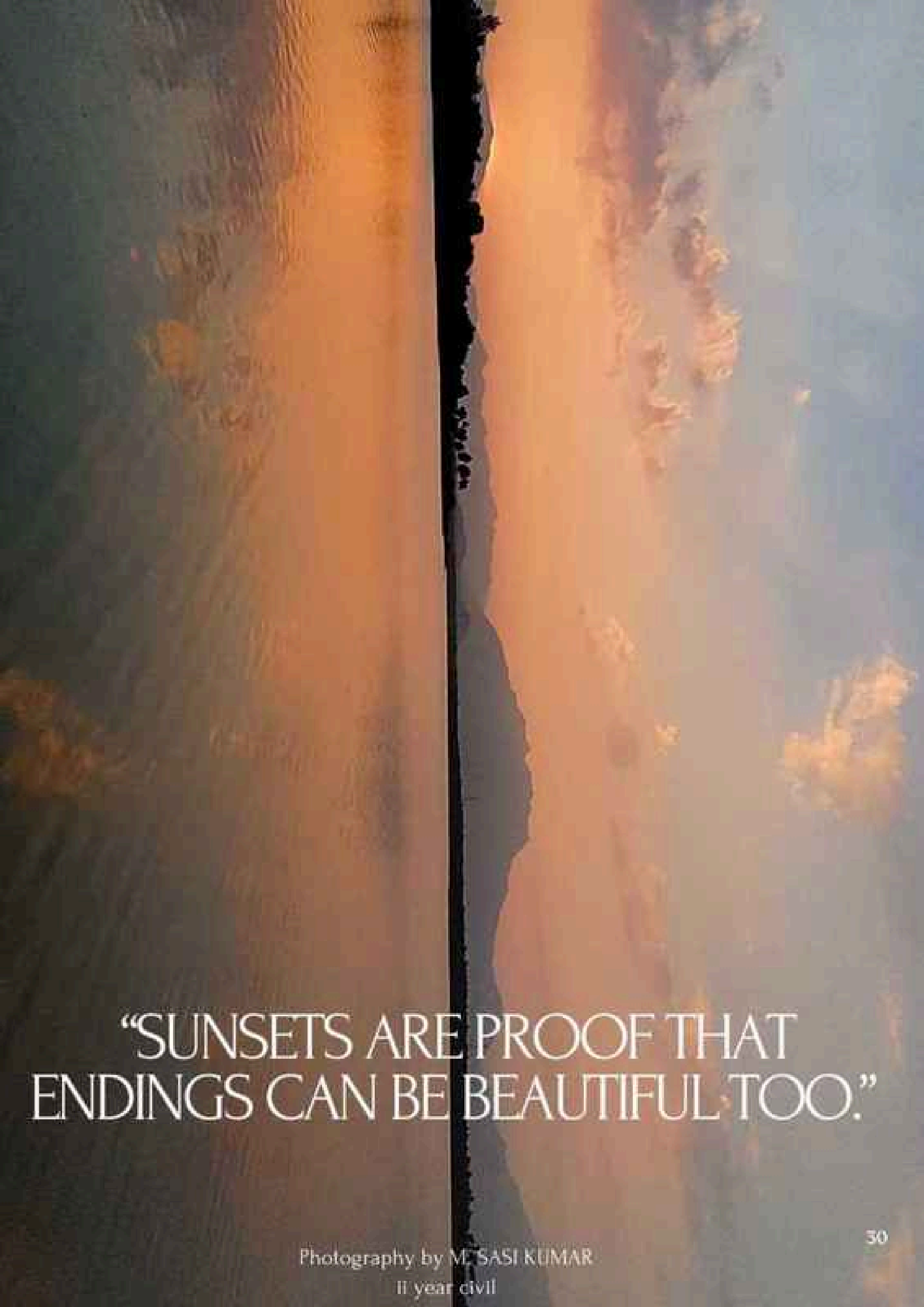



Arun Lakesh

1-year civil

photo graphy





“SUNSETS ARE PROOF THAT
ENDINGS CAN BE BEAUTIFUL TOO.”



"THOUSANDS HAVE LIVED
WITHOUT LOVE. NOT ONE
WITHOUT WATER."
— W. H. AUDEN, POET.

CLOUDS COME FLOATING INTO MY LIFE, NO
LONGER TO CARRY RAIN OR USHER STORM,
BUT TO ADD COLOR TO MY SUNSET SKY.....



PHOTOGRAPHY BY
B. PRAVEEN
II YEAR CIVIL



"Life is a journey, and college is the start of a great adventure".

"One child, one teacher, one book, one pen can change the world."

malala yousafzai.

**Photography by
D.C.Esakki Hari Ganesh
II YEAR CIVIL**



"THE EARTH HAS MUSIC
FOR THOSE WHO LISTEN."
— SHAKESPEARE

PHOTOGRAPHY BY
T. SURYA
II YEAR CIVIL



PHOTOGRAPHY
BY
P.THANA BALA
II YEAR CIVIL





*Thank
you!*